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AssignmentNumber:13.4 (Present assignment number)/24(Total number of assignments)

Q.No.	Question	ExpectedTime to complete
1	Lab 13: Code Refactoring – Improving Legacy Code with AI Suggestions Lab Objectives • Identify code smells and inefficiencies in legacy Python scripts. • Use AI-assisted coding tools to refactor for readability, maintainability, and performance. • Apply modern Python best practices while ensuring output correctness. Learning Outcome After completing this assignment, students will be able to: • Identify refactoring opportunities in legacy Python code • Apply AI-assisted suggestions to refactor code without changing behavior • Refactor Python code using Pythonic and scalable constructs • Critically evaluate AI-generated refactoring recommendations	Week 7
	Task 1: Refactoring Data Transformation Logic Scenario You are maintaining a data preprocessing script where numerical transformations are written using verbose loops. Task Description • Review the legacy code that computes transformed values using an explicit loop. • Use an AI tool to suggest a more Pythonic refactoring approach.	

	Refactor the code using list comprehensions or helper functions while preserving the output. Legacy Code <pre>values = [2, 4, 6, 8, 10] doubled = [] for v in values: doubled.append(v * 2) print(doubled)</pre> Expected Output <pre>[4, 8, 12, 16, 20]</pre> Prompt: I have legacy Python code that uses a loop to transform data. Code: <pre>values = [2, 4, 6, 8, 10] doubled = [] for v in values: doubled.append(v * 2)</pre> Refactor this code using a more Pythonic approach like list comprehension or helper functions. Keep the output exactly the same. Explain why the refactored version is better in terms of readability and performance.	
	Task 2: Improving Text Processing Code Readability Scenario You are working on a log message generator that builds messages word by word. Task Description - Analyze the legacy code that constructs a sentence using repeated string concatenation. - Ask AI to suggest a more efficient and readable approach.	

	Refactor the code accordingly while keeping the final output unchanged. Legacy Code <pre>words = ["Refactoring", "with", "AI", "improves", "quality"] message = "" for w in words: message += w + " " print(message.strip())</pre> Expected Output Refactoring with AI improves quality Prompt: I have legacy Python code that builds a sentence using string concatenation inside a loop. Code: <pre>words = ["Refactoring", "with", "AI", "improves", "quality"] message = "" for w in words: message += w + " "</pre> Refactor this code into a cleaner and more efficient approach. Do not change the final output. Explain why your method is more efficient and readable.	
	Task 3: Safer Access to Configuration Data Scenario You are maintaining a configuration loader where dictionary keys may or may not exist depending on deployment settings. Task Description - Review the legacy code that manually checks for dictionary keys. - Use AI suggestions to refactor the code using safer dictionary access methods. - Ensure the behavior remains the same for missing keys.	

	Legacy Code <pre>config = {"host": "localhost", "port": 8080} if "timeout" in config: print(config["timeout"]) else: print("Default timeout used")</pre> Expected Output Default timeout used Prompt: I have Python code that manually checks if a dictionary key exists. Code: <pre>config = {"host": "localhost", "port": 8080} if "timeout" in config: print(config["timeout"]) else: print("Default timeout used")</pre> Refactor this code using safer dictionary access methods (like get()). Ensure the behavior remains the same when keys are missing. Explain why the new approach is safer.	
	Task 4: Refactoring Conditional Logic for Scalability Scenario You are enhancing a utility module that performs different operations based on user input. Task Description - Examine the multiple if–elif conditions used to determine operations. - Ask AI to suggest a cleaner, scalable alternative.	

- Refactor the logic using mapping techniques while preserving functionality.

Legacy Code

```
action = "divide"
x, y = 10, 2

if action == "add":
    result = x + y
elif action == "subtract":
    result = x - y
elif action == "multiply":
    result = x * y
elif action == "divide":
    result = x / y
else:
    result = None

print(result)
```

Expected Output

5.0

Prompt:

I have Python code using multiple if-elif conditions to perform operations.

Code:

```
action = "divide"
x, y = 10, 2

if action == "add":
    result = x + y
elif action == "subtract":
    result = x - y
elif action == "multiply":
    result = x * y
elif action == "divide":
    result = x / y
else:
    result = None
```

Refactor this using a scalable and cleaner approach such as mapping or dictionaries.
Keep functionality unchanged.
Explain why the new design is better for scalability.

Task 5: Simplifying Search Logic in Collections

Scenario

You are reviewing **legacy inventory-check code** that uses manual loops to find elements.

Task Description

- Identify the explicit loop used for searching an item.
- Use AI assistance to refactor the logic into a more concise and readable form.
- Maintain the same output behavior.

Legacy Code

```
inventory = ["pen", "notebook", "eraser", "marker"]
found = False

for item in inventory:
    if item == "eraser":
        found = True
        break

print("Item Available" if found else "Item Not Available")
```

Expected Output

Item Available

Prompt:

I have legacy Python code that searches for an item using a manual loop.

Code:

```
inventory = ["pen", "notebook", "eraser", "marker"]
found = False

for item in inventory:
    if item == "eraser":
        found = True
        break
```

	Refactor this code into a shorter and more Pythonic solution. Do not change the output behavior. Explain why your solution improves readability.	
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